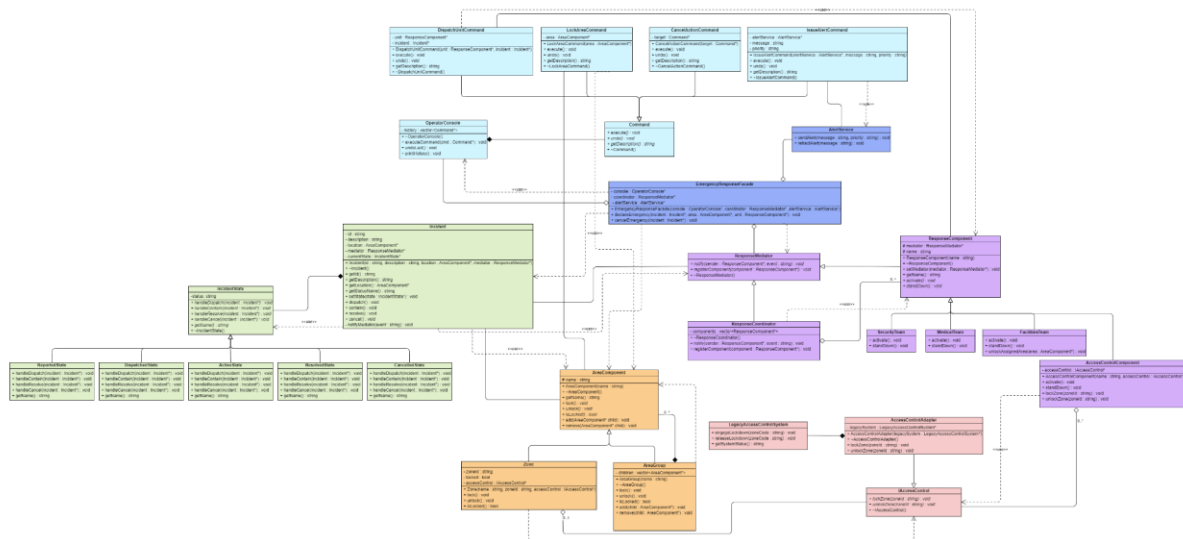


COS 214 - Practical 2

Team members

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Task 1: Analyse the Scenario and Design the System



(See image prac5_ClassUML.png for a better view)

System description

CampusGuard coordinates a university's emergency response from the moment an incident is reported until it's resolved. It tracks each incident's lifecycle, dispatches security, medical and facilities teams, manages building access (individual rooms or whole buildings) during a response, integrates with a legacy access-control system, and gives operators a single point of entry for declaring or cancelling an emergency rather than juggling every subsystem by hand.

GoF participants

Command

Command is the abstract Command, with DispatchUnitCommand, LockAreaCommand, IssueAlertCommand and CancelActionCommand as concrete commands. OperatorConsole is the invoker (holds the history vector, exposes executeCommand/undoLast). Receivers vary per command: ResponseComponent + Incident for DispatchUnitCommand, AreaComponent for LockAreaCommand, AlertService for IssueAlertCommand, and —

interestingly — another Command for `CancelActionCommand`, which wraps and reverses a previously-executed command rather than acting on a domain object directly.

Mediator

`ResponseMediator` is the Mediator interface (`notify`, `registerComponent`), implemented by `ResponseCoordinator`. `ResponseComponent` is the Colleague base (holds a mediator pointer, exposes `setMediator`), with `SecurityTeam`, `MedicalTeam` and `FacilitiesTeam` as concrete colleagues.

Adapter

`IAccessControl` is the Target interface (`lockZone/unlockZone`), `AccessControlAdapter` is the Adapter, and `LegacyAccessControlSystem` is the Adaptee, exposing a genuinely different interface (`engageLockdown/releaseLockdown`, keyed by `zoneCode` rather than `zoneld`). `Zone` and `FacilitiesTeam` are the clients — both depend only on `IAccessControl` and never see the legacy system or the adapter itself.

Facade

`EmergencyResponseFacade` is the Facade, holding references to the three subsystems it orchestrates (`OperatorConsole`, `ResponseMediator`, `AlertService`). Its two operations, `declareEmergency(...)` and `cancelEmergency(...)`, are the higher-level entry points; `Incident`, `AreaComponent` and `ResponseComponent` are passed in as parameters rather than stored, since the facade coordinates them per-call rather than owning them.

The two additional patterns

State

(`IncidentState` / `Incident`) — `Incident` holds a `currentState` and delegates `dispatch()`, `contain()`, `resolve()`, `cancel()` to it, with `ReportedState`, `ActiveState/DispatchedState`, `ResolvedState` and `CancelledState` as concrete states. This solves the problem of legal-transition checking without a growing if/switch chain inside `Incident` itself, and gives a natural place for an invalid transition (e.g. resolving an incident that was never dispatched) to fail sensibly rather than silently.

Composite

(`AreaComponent` / `Zone` / `AreaGroup`) — `AreaComponent` is the component, `Zone` the leaf (owns an `IAccessControl*`, tracks its own locked state), `AreaGroup` the composite (holds a `vector<AreaComponent*>` and recurses `lock()/unlock()` over its children). This means `LockAreaCommand` and `FacilitiesTeam::unlockAssignedArea` can treat a single room and an entire building identically, with no type-checking required.

Associations, multiplicities, ownership

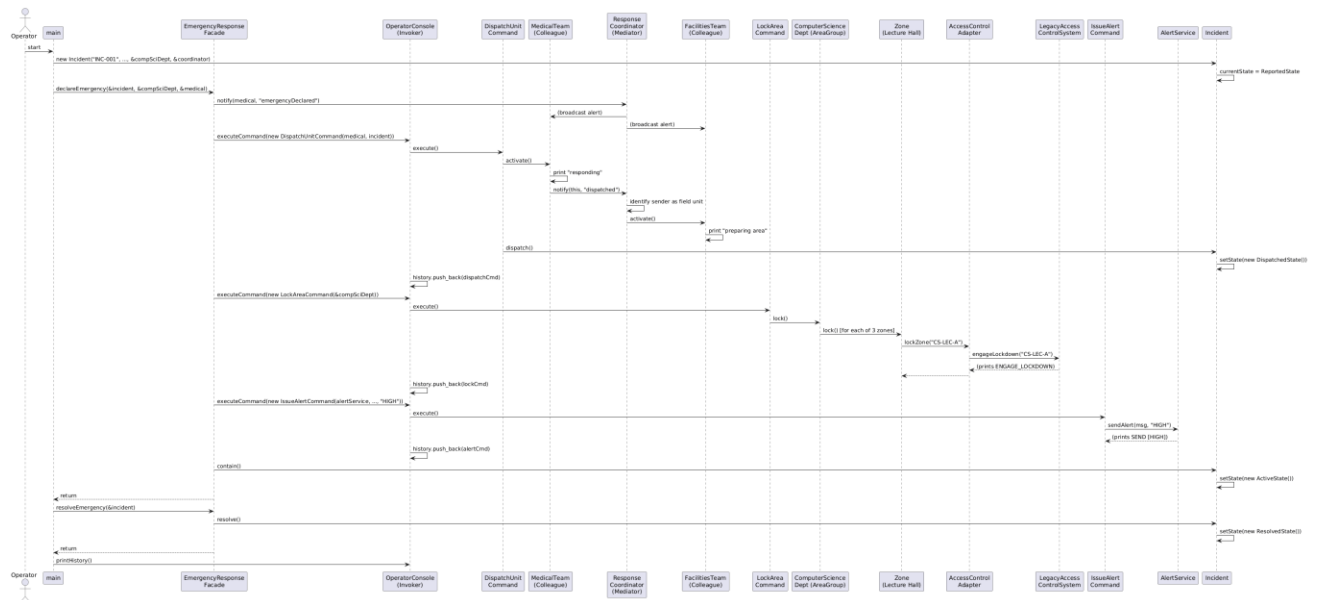
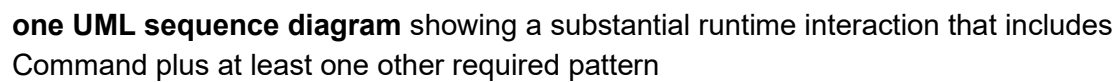
- `Incident` *-- 1 `IncidentState` — composition. `Incident` owns and swaps its current state on each transition.

- Incident --> 0..1 AreaComponent, Incident --> 0..1 ResponseMediator — plain associations, not owned by Incident.
- OperatorConsole *-- 0..* Command — composition. The console's history is why it declares a destructor; it's responsible for deleting every command it ever executed.
- DispatchUnitCommand --> ResponseComponent, Incident; LockAreaCommand --> AreaComponent; IssueAlertCommand --> AlertService; CancelActionCommand --> Command — all plain associations (receivers held but not owned).
- ResponseCoordinator o-- 0..* ResponseComponent — aggregation. Colleagues are shared with Command as receivers, so the coordinator can't be the sole owner.
- AccessControlAdapter *-- 1 LegacyAccessControlSystem — composition. Nothing else references the legacy system, so the adapter owns its lifetime.
- Zone o-- 1 IAccessControl — aggregation. The same IAccessControl instance is shared wherever access control is needed; Zone doesn't create it.
- AreaGroup *-- 0..* AreaComponent — composition. Textbook Composite: deleting a building deletes its rooms.
- EmergencyResponseFacade o-- OperatorConsole, ResponseMediator, AlertService — aggregation. Per the brief's requirement that subsystem operations "remain independently usable," the facade can't own these outright.

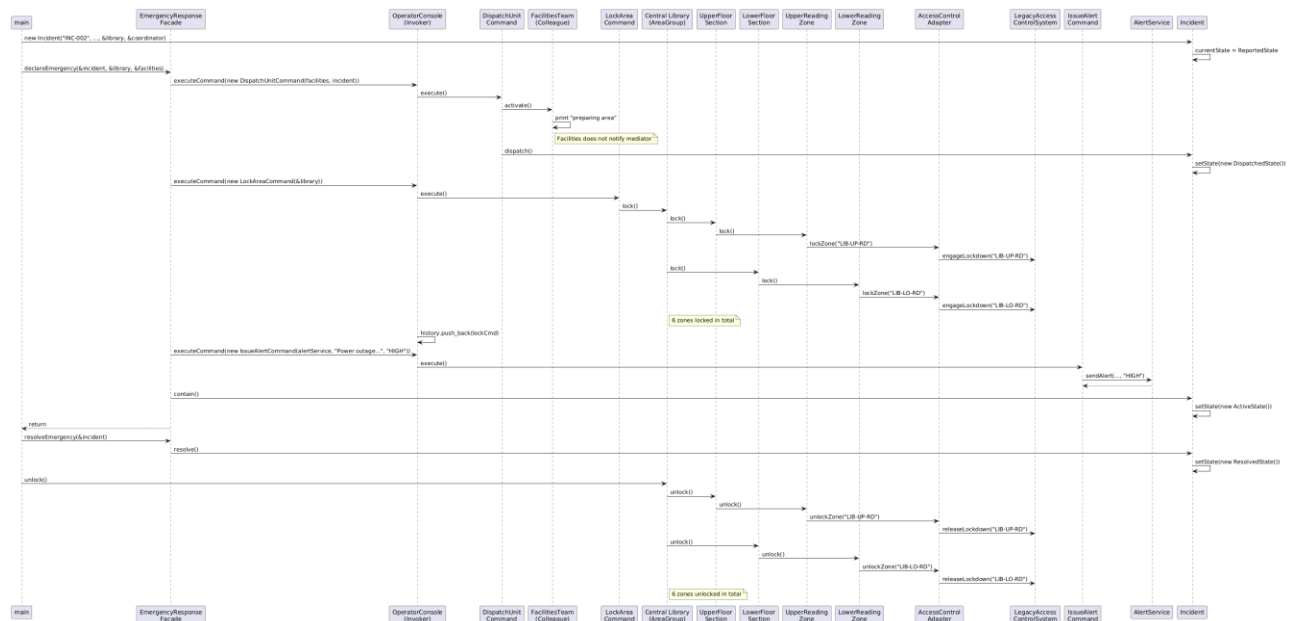
Where the six patterns meet

A single `declareEmergency()` call demonstrates the integration: the Facade builds a `DispatchUnitCommand` and runs it through the `OperatorConsole` (Command); executing it calls `incident->dispatch()`, which the current `IncidentState` handles, transitioning the incident (State); the dispatched `ResponseComponent` reports itself to `ResponseCoordinator` (Mediator), which directs a `FacilitiesTeam` to prepare the area; `FacilitiesTeam::unlockAssignedArea()` calls `unlock()` on whatever `AreaComponent` it's given — a single `Zone` or an entire `AreaGroup` (Composite) — and that `Zone` reaches `IAccessControl`, actually implemented by `AccessControlAdapter`, which translates the call for `LegacyAccessControlSystem` (Adapter). Command names the action, State governs what's legal, Mediator decides who else needs to react, Composite makes area operations uniform regardless of scale, and Adapter hides the legacy system underneath — with Facade sitting above all of it, sequencing without doing the work itself.

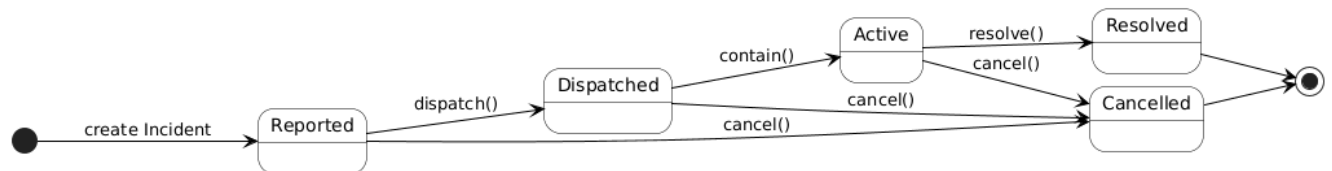
one complete UML class diagram for the final system



one second UML sequence diagram showing a different workflow that includes the Facade or Adapter;



one additional behavioural diagram of your choice (activity, state or sequence) that best explains one of your two chosen patterns.



Dockerfile, docker-compose.yml, and the Docker run instructions

- Run instructions:
 - 1.Start Docker daemon
sudo service docker start
 - 2.Go to the project directory
cd ~/COS214/COS214_Practical_5-main
 - 3.build and run program
sudo docker compose up --build
- Docker Results:

ameltonkhoza@MnadlaKhoza:~/COS214/COS214_Practical_5-main\$ sudo docker compose up --build

WARN[0000] Docker Compose is configured to build using Bake, but buildx isn't installed

[+] Building 189.8s (16/16) FINISHED
docker:default

=> [campusguard internal] load build definition from Dockerfile
0.2s

=> => transferring dockerfile: 541B
0.0s

=> [campusguard internal] load metadata for docker.io/library/ubuntu:22.04
2.8s

=> [campusguard internal] load .dockerignore
0.1s

=> => transferring context: 384B
0.0s

=> [campusguard internal] load build context
0.2s

=> => transferring context: 77.42kB
0.0s

=> [campusguard builder 1/7] FROM
docker.io/library/ubuntu:22.04@sha256:b8b6ee6aa931ecd9d0d952abc34dc0e5f7c6a30c6b
b71b079fe399fde0329c02 10.0s

=> => resolve
docker.io/library/ubuntu:22.04@sha256:b8b6ee6aa931ecd9d0d952abc34dc0e5f7c6a30c6b
b71b079fe399fde0329c02 0.1s

=> => sha256:20c3783cc497b5b0df1fc5f92bd64c3d2fbb24057c88692c8fda205b6ea8a2f2
29.75MB / 29.75MB
8.4s

=> => extracting
sha256:20c3783cc497b5b0df1fc5f92bd64c3d2fbb24057c88692c8fda205b6ea8a2f2
1.3s

=> [campusguard stage-1 2/4] RUN apt-get update && apt-get install -y gdb valgrind
&& rm -rf /var/lib/apt/lists/* 130.2s

=> [campusguard builder 2/7] RUN apt-get update && apt-get install -y g++ make
&& rm -rf /var/lib/apt/lists/* 157.0s

=> [campusguard stage-1 3/4] WORKDIR /app
0.1s

=> [campusguard builder 3/7] WORKDIR /build
0.1s

=> [campusguard builder 4/7] COPY Makefile ./
0.1s

=> [campusguard builder 5/7] COPY *.cpp ./
0.1s

=> [campusguard builder 6/7] COPY *.h ./
0.1s

=> [campusguard builder 7/7] RUN make clean && make
6.5s

=> [campusguard stage-1 4/4] COPY --from=builder /build/campusguard /app/campusguard
0.1s

=> [campusguard] exporting to image
12.3s

=> => exporting layers
9.9s

=> => exporting manifest
sha256:058193793ba3fdbcc62b5ecf4e9e180a011602c7f92950207a8a218d132d907b
0.0s

=> => exporting config
sha256:51a9c83e6e7decb8bf21371e2a8f00e6d42091cf355f003f01e14ea38509c4cd
0.0s

=> => exporting attestation manifest
sha256:84b7551656caadebf386fd00d1dbb17e066d430843f5af5fa09de1f0ffbc3dfd
0.0s

=> => exporting manifest list
sha256:f2b15c79c42378395f72d225aac7d0384678bc40de9c6135e733477704410eef
0.0s

=> => naming to docker.io/library/campusguard:latest
0.0s

=> => unpacking to docker.io/library/campusguard:latest
2.2s

=> [campusguard] resolving provenance for metadata file
0.0s

[+] Running 3/3

✓ campusguard Built
0.0s

✓ Network cos214_practical_5-main_default Created
0.0s

✓ Container campusguard Created
0.4s

Attaching to campusguard

campusguard | CampusGuard: Emergency Response Coordination

campusguard | [Coordinator] Registered Campus Security as a response colleague.

campusguard | [Coordinator] Registered Medical Response as a response colleague.

campusguard | [Coordinator] Registered Facilities Crew as a response colleague.

campusguard |

campusguard |

campusguard | SCENARIO 1: Medical emergency during the COS 132 lecture (Department of Computer Science)

campusguard |

campusguard |

campusguard | [Main] Incident INC-001 created. Status = Reported

campusguard | [Coordinator] Medical Response reported event: "emergencyDeclared"

campusguard | [Coordinator] Emergency declared by Medical Response. Notifying all colleagues.

campusguard | [Coordinator] Alerting Campus Security

campusguard | [Coordinator] Alerting Facilities Crew

campusguard | [MedicalTeam] Medical Response responding to the incident location.

campusguard | [Coordinator] Medical Response reported event: "dispatched"

campusguard | [Coordinator] Directing Facilities Crew to prepare the affected area.

campusguard | [FacilitiesTeam] Facilities Crew preparing the affected area (unlocking doors, clearing routes).

campusguard | [State] Reported -> Dispatched

campusguard | [Incident] INC-001 -> Dispatched

campusguard | [Incident] state event: "Dispatched" (facade will mediate on our behalf)

campusguard | [AccessControlAdapter] lockZone("CS-LEC-A") -> engageLockdown("CS-LEC-A")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code CS-LEC-A

campusguard | [AccessControlAdapter] lockZone("CS-CLAB") -> engageLockdown("CS-CLAB")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code CS-CLAB

campusguard | [AccessControlAdapter] lockZone("CS-SRV") -> engageLockdown("CS-SRV")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code CS-SRV

campusguard | [AlertService] SEND [HIGH] Emergency reported: Student collapsed during COS 132 lecture in Lecture Hall A

campusguard | [State] Dispatched -> Active

campusguard | [Incident] INC-001 -> Active

campusguard | [Incident] state event: "Active" (facade will mediate on our behalf)

campusguard |

campusguard | [Main] After declareEmergency. Status = Active

campusguard | [State] Active -> Resolved

campusguard | [Incident] INC-001 -> Resolved

campusguard | [Incident] state event: "Resolved" (facade will mediate on our behalf)

campusguard | [Main] After resolveEmergency. Status = Resolved

campusguard |

campusguard | [Main] Command history so far:

campusguard | Dispatch: Medical Response to incident: Student collapsed during COS 132 lecture in Lecture Hall A

campusguard | Department of Computer Science is Locked

campusguard | Alert: Emergency reported: Student collapsed during COS 132 lecture in Lecture Hall A | Priority: HIGH

campusguard |

campusguard |

campusguard | SCENARIO 2: Electricity outage at the Library

campusguard |

campusguard | [Main Incident] Incident INC-002 created, Status = Reported

campusguard | [Coordinator] Facilities Crew reported event: "emergencyDeclared"

campusguard | [Coordinator] Emergency declared by Facilities Crew. Notifying all colleagues.

campusguard | [Coordinator] Alerting Campus Security

campusguard | [Coordinator] Alerting Medical Response

campusguard | [FacilitiesTeam] Facilities Crew preparing the affected area (unlocking doors, clearing routes).

campusguard | [State] Reported -> Dispatched

campusguard | [Incident] INC-002 -> Dispatched

campusguard | [Incident] state event: "Dispatched" (facade will mediate on our behalf)

campusguard | [AccessControlAdapter] lockZone("LIB-UP-RD") -> engageLockdown("LIB-UP-RD")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-UP-RD

campusguard | [AccessControlAdapter] lockZone("LIB-UP-SIL") -> engageLockdown("LIB-UP-SIL")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-UP-SIL

campusguard | [AccessControlAdapter] lockZone("LIB-UP-GRP") -> engageLockdown("LIB-UP-GRP")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-UP-GRP

campusguard | [AccessControlAdapter] lockZone("LIB-LO-RD") -> engageLockdown("LIB-LO-RD")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-LO-RD

campusguard | [AccessControlAdapter] lockZone("LIB-LO-CMN") -> engageLockdown("LIB-LO-CMN")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-LO-CMN

campusguard | [AccessControlAdapter] lockZone("LIB-LO-CUB") -> engageLockdown("LIB-LO-CUB")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-LO-CUB

campusguard | [AlertService] SEND [HIGH] Emergency reported: Electricity outage - Central Library must be evacuated

campusguard | [State] Dispatched -> Active

campusguard | [Incident] INC-002 -> Active

campusguard | [Incident] state event: "Active" (facade will mediate on our behalf)

campusguard |

campusguard | [Main Incident] After declareEmergency. Status = Active

campusguard | [Main Incident] Library is now locked for evacuation.

campusguard |

campusguard | [Main Incident] Power restored - resolving incident and unlocking the library.

campusguard | [State] Active -> Resolved

campusguard | [Incident] INC-002 -> Resolved

campusguard | [Incident] state event: "Resolved" (facade will mediate on our behalf)

campusguard | [AccessControlAdapter] unlockZone("LIB-UP-RD") ->
releaseLockdown("LIB-UP-RD")

campusguard | [LegacySystem] RELEASE_LOCKDOWN signal sent for zone code LIB-UP-RD

campusguard | [AccessControlAdapter] unlockZone("LIB-UP-SIL") ->
releaseLockdown("LIB-UP-SIL")

campusguard | [LegacySystem] RELEASE_LOCKDOWN signal sent for zone code LIB-UP-SIL

campusguard | [AccessControlAdapter] unlockZone("LIB-UP-GRP") ->
releaseLockdown("LIB-UP-GRP")

campusguard | [LegacySystem] RELEASE_LOCKDOWN signal sent for zone code LIB-UP-GRP

campusguard | [AccessControlAdapter] unlockZone("LIB-LO-RD") ->
releaseLockdown("LIB-LO-RD")

campusguard | [LegacySystem] RELEASE_LOCKDOWN signal sent for zone code LIB-LO-RD

campusguard | [AccessControlAdapter] unlockZone("LIB-LO-CMN") ->
releaseLockdown("LIB-LO-CMN")

campusguard | [LegacySystem] RELEASE_LOCKDOWN signal sent for zone code LIB-LO-CMN

campusguard | [AccessControlAdapter] unlockZone("LIB-LO-CUB") ->
releaseLockdown("LIB-LO-CUB")

campusguard | [LegacySystem] RELEASE_LOCKDOWN signal sent for zone code LIB-LO-CUB

campusguard | [Main Incident] After resolveEmergency and unlock. Status = Resolved

campusguard |

campusguard | [Main Incident] Command history after power outage scenario:

campusguard | Dispatch: Medical Response to incident: Student collapsed during COS 132 lecture in Lecture Hall A

campusguard | Department of Computer Science is Locked

campusguard | Alert: Emergency reported: Student collapsed during COS 132 lecture in Lecture Hall A | Priority: HIGH

campusguard | Dispatch: Facilities Crew to incident: Electricity outage - Central Library must be evacuated

campusguard | Central Library is Unlocked

campusguard | Alert: Emergency reported: Electricity outage - Central Library must be evacuated | Priority: HIGH

campusguard |

campusguard |

campusguard | SCENARIO 3: Natural disaster - campus-wide lockdown

campusguard |

campusguard | [Main Incident] Incident INC-003 created. Status = Reported

campusguard | [Coordinator] Campus Security reported event: "emergencyDeclared"

campusguard | [Coordinator] Emergency declared by Campus Security. Notifying all colleagues.

campusguard | [Coordinator] Alerting Medical Response

campusguard | [Coordinator] Alerting Facilities Crew

campusguard | [SecurityTeam] Campus Security mobilising to the incident location.

campusguard | [Coordinator] Campus Security reported event: "dispatched"

campusguard | [Coordinator] Directing Facilities Crew to prepare the affected area.

campusguard | [FacilitiesTeam] Facilities Crew preparing the affected area (unlocking doors, clearing routes).

campusguard | [State] Reported -> Dispatched

campusguard | [Incident] INC-003 -> Dispatched

campusguard | [Incident] state event: "Dispatched" (facade will mediate on our behalf)

campusguard | [AccessControlAdapter] lockZone("EBIT-HR") -> engageLockdown("EBIT-HR")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code EBIT-HR

campusguard | [AccessControlAdapter] lockZone("IS-LEC-B") -> engageLockdown("IS-LEC-B")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code IS-LEC-B

campusguard | [AccessControlAdapter] lockZone("IS-DATA") -> engageLockdown("IS-DATA")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code IS-DATA

campusguard | [AccessControlAdapter] lockZone("IS-MEDIA") -> engageLockdown("IS-MEDIA")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code IS-MEDIA

campusguard | [AccessControlAdapter] lockZone("LIB-UP-RD") -> engageLockdown("LIB-UP-RD")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-UP-RD

campusguard | [AccessControlAdapter] lockZone("LIB-UP-SIL") -> engageLockdown("LIB-UP-SIL")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-UP-SIL

campusguard | [AccessControlAdapter] lockZone("LIB-UP-GRP") -> engageLockdown("LIB-UP-GRP")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-UP-GRP

campusguard | [AccessControlAdapter] lockZone("LIB-LO-RD") -> engageLockdown("LIB-LO-RD")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-LO-RD

campusguard | [AccessControlAdapter] lockZone("LIB-LO-CMN") -> engageLockdown("LIB-LO-CMN")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-LO-CMN

campusguard | [AccessControlAdapter] lockZone("LIB-LO-CUB") -> engageLockdown("LIB-LO-CUB")

campusguard | [LegacySystem] ENGAGE_LOCKDOWN signal sent for zone code LIB-LO-CUB

campusguard | [AlertService] SEND [HIGH] Emergency reported: Tornado warning - full campus lockdown

campusguard | [State] Dispatched -> Active

campusguard | [Incident] INC-003 -> Active

campusguard | [Incident] state event: "Active" (facade will mediate on our behalf)

campusguard |

campusguard | [Main Incident] Deploying Medical and Facilities as backup:

campusguard | [Coordinator] Medical Response reported event: "emergencyDeclared"

campusguard | [Coordinator] Emergency declared by Medical Response. Notifying all colleagues.

campusguard | [Coordinator] Alerting Campus Security

campusguard | [Coordinator] Alerting Facilities Crew

campusguard | [MedicalTeam] Medical Response responding to the incident location.

campusguard | [Coordinator] Medical Response reported event: "dispatched"

campusguard | [Coordinator] Directing Facilities Crew to prepare the affected area.

campusguard | [FacilitiesTeam] Facilities Crew preparing the affected area (unlocking doors, clearing routes).

campusguard | [AlertService] SEND [HIGH] Emergency reported: Tornado warning - full campus lockdown

campusguard | [Coordinator] Facilities Crew reported event: "emergencyDeclared"

campusguard | [Coordinator] Emergency declared by Facilities Crew. Notifying all colleagues.

campusguard | [Coordinator] Alerting Campus Security

campusguard | [Coordinator] Alerting Medical Response

campusguard | [FacilitiesTeam] Facilities Crew preparing the affected area (unlocking doors, clearing routes).

campusguard | [AlertService] SEND [HIGH] Emergency reported: Tornado warning - full campus lockdown

campusguard |

campusguard | [Main Incident] After full deployment. Status = Active

campusguard | [State] Active -> Resolved

campusguard | [Incident] INC-003 -> Resolved

campusguard | [Incident] state event: "Resolved" (facade will mediate on our behalf)

campusguard | [Main Incident] After resolveEmergency. Status = Resolved

campusguard |

campusguard | [Main Incident] Full command history:

campusguard | Dispatch: Medical Response to incident: Student collapsed during COS 132 lecture in Lecture Hall A

campusguard | Department of Computer Science is Locked

campusguard | Alert: Emergency reported: Student collapsed during COS 132 lecture in Lecture Hall A | Priority: HIGH

campusguard | Dispatch: Facilities Crew to incident: Electricity outage - Central Library must be evacuated

campusguard | Central Library is Locked

campusguard | Alert: Emergency reported: Electricity outage - Central Library must be evacuated | Priority: HIGH

campusguard | Dispatch: Campus Security to incident: Tornado warning - full campus lockdown

campusguard | University of Pretoria is Locked

campusguard | Alert: Emergency reported: Tornado warning - full campus lockdown | Priority: HIGH

campusguard | Dispatch: Medical Response to incident: Tornado warning - full campus lockdown

campusguard | University of Pretoria is Locked

campusguard | Alert: Emergency reported: Tornado warning - full campus lockdown | Priority: HIGH

campusguard | Dispatch: Facilities Crew to incident: Tornado warning - full campus lockdown

campusguard | University of Pretoria is Locked

campusguard | Alert: Emergency reported: Tornado warning - full campus lockdown | Priority: HIGH

campusguard |

campusguard |

campusguard | Cross-pattern checkpoints

campusguard | Composite : institution.isLocked() = true

campusguard | Composite : ebitFaculty.isLocked() = true

campusguard | Composite : facultyHROffice.isLocked() = true

campusguard | Composite : compSciDept.isLocked() = true

campusguard | Composite : csLectureHall.isLocked() = true

campusguard | Composite : csComputerLab.isLocked() = true

campusguard | Composite : csServerRoom.isLocked() = true

campusguard | Composite : infoSciDept.isLocked() = true

campusguard | Composite : isLectureHall.isLocked() = true

campusguard | Composite : isDataLab.isLocked() = true

campusguard | Composite : isMultimedia.isLocked() = true

campusguard | Composite : library.isLocked() = true

campusguard | Composite : upperFloorSection.isLocked() = true

campusguard | Composite : lowerFloorSection.isLocked() = true

campusguard | Composite : upperReadingZone.isLocked() = true

campusguard | Composite : upperSilentStudyZone.isLocked() = true

campusguard | Composite : upperGroupStudyZone.isLocked() = true

campusguard | Composite : lowerReadingZone.isLocked() = true

campusguard | Composite : commonStudyArea.isLocked() = true

campusguard | Composite : cubicles.isLocked() = true

campusguard | Shutdown complete

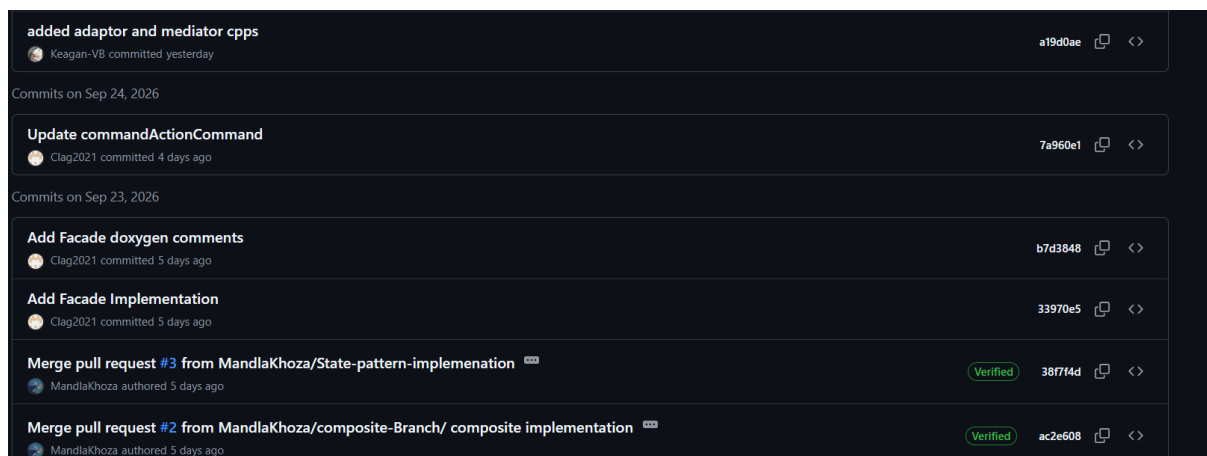
campusguard exited with code 0

ameltonkhoza@MnadlaKhoza:~/COS214/COS214_Practical_5-main\$

GitHub repository link and collaboration reflection

https://github.com/MandlaKhoza/COS214_Practical_5

The work collaboration on github was done so: a branch was created for just the header files before a branch for every pattern was created where each pattern was to be implemented separately. Once each pattern was completed it was merged into the main branch where the final testing main and demo main was completed. This separation allowed us to work independent of each other and avoid any merge conflicts that could occur if there were multiple coders working on the same branch. This also prevented any unnecessary or harmful code from being pushed onto the main branch which saved us time with troubleshooting problems that arose during testing.



Contribution statement for each team member

We all made a short mockup of what the system could look like before proceeding with a meeting to decide on the final system. Keagan worked on making a class uml of the entire system before we divided the code between the 3 of us. Amelton worked on implementing the composite, state and testing main. Keagan worked on the mediator, adapter, and the demo main. Nontokoza implemented the facade and the command.

GDB investigation:

- **Before investigation:**

Breakpoint 1 at 0xa61b: file ResponseCoordinator.cpp, line 83.

Starting program: /home/ameltonkhoza/COS214/COS214_Practical_5-main/campusguard

This GDB supports auto-downloading debuginfo from the following URLs:

<<https://debuginfod.ubuntu.com>>

Debuginfod has been disabled.

To make this setting permanent, add 'set debuginfod enabled off' to .gdbinit.

[Thread debugging using libthread_db enabled]

Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, ResponseCoordinator::notify (this=0x7ffffffd120, sender=0x7ffffffd1b0, event="EMERGENCY DECLARED: ") at ResponseCoordinator.cpp:83

83 {

\$1 = "EMERGENCY DECLARED: " //emergency declared different from teh required function

\$2 = (ResponseComponent *) 0x7ffffffd1b0

84 if (sender == nullptr) {

91 << " reported event: \"" << event << "\"" << std::endl;

90 std::cout << "[Coordinator] " << sender->getName()

91 << " reported event: \"" << event << "\"" << std::endl;

90 std::cout << "[Coordinator] " << sender->getName()

98 if (event == "dispatched") {

123 else if (event == "standDown") {

141 else if (event == "emergencyDeclared") {

152 else if (event == "emergencyCancelled") {

163 << event << "\"." << std::endl;

165 }

EmergencyResponseFacade::declareEmergency (this=0x7ffffffd100, incident=0x7ffffffd8b0, area=0x7ffffffd250, unit=0x7ffffffd1b0) at EmergencyResponseFacade.cpp:54

54 DispatchUnitCommand* dispatch = new DispatchUnitCommand(unit, incident);

(gdb) quit

A debugging session is active.

Inferior 1 [process 97029] will be killed.

Quit anyway? (y or n) y

//This is before fixing teh main to make sure that we accomodate teh correct event.

- **After investigation:**

Breakpoint 1 at 0xa61b: file ResponseCoordinator.cpp, line 83.

Starting program: /home/ameltonkhoza/COS214/COS214_Practical_5-main/campusguard

This GDB supports auto-downloading debuginfo from the following URLs:

<<https://debuginfod.ubuntu.com>>

Debuginfod has been disabled.

To make this setting permanent, add 'set debuginfod enabled off' to .gdbinit.

[Thread debugging using libthread_db enabled]

Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, ResponseCoordinator::notify (this=0x7ffffffd120, sender=0x7ffffffd1b0, event="emergencyDeclared") at ResponseCoordinator.cpp:83

```
83 {  
$1 = "emergencyDeclared"  
$2 = (ResponseComponent *) 0x7ffffffd1b0  
84   if (sender == nullptr && event != "emergencyCancelled") {  
90       << " reported event: \"" << event << "\"" << std::endl;  
89   std::cout << "[Coordinator] " << sender->getName()  
90       << " reported event: \"" << event << "\"" << std::endl;  
89   std::cout << "[Coordinator] " << sender->getName()  
97   if (event == "dispatched") {  
122       else if (event == "standDown") {  
140       else if (event == "emergencyDeclared") { //we no longer pass this section  
142           << sender->getName() << ". Notifying all colleagues."  
143           << std::endl;  
142           << sender->getName() << ". Notifying all colleagues."  
144           for (ResponseComponent* c : components) {  
145               if (c != sender) {
```

Continuing.

Breakpoint 1, ResponseCoordinator::notify (this=0x7ffffffd120, sender=0x7ffffffd1b0, event="dispatched") at ResponseCoordinator.cpp:83

83 {

(gdb) quit

A debugging session is active.

Inferior 1 [process 101443] will be killed.

Quit anyway? (y or n) y

ameltonkhoza@MnadlaKhoza:~/COS214/COS214_Practical_5-main\$ ^C

ameltonkhoza@MnadlaKhoza:~/COS214/COS214_Practical_5-main\$

//now scenaion 1 has access to the emergency decalred if statement.

Valgrind Evidence:

==92340== Memcheck, a memory error detector

==92340== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.

==92340== Using Valgrind-3.26.0-218cee2feb-20251024 and LibVEX; rerun with -h for copyright info

==92340== Command: ./campusguard

==92340== Parent PID: 91985

==92340==

--92340--

--92340-- Valgrind options:

--92340-- --leak-check=full

--92340-- --show-leak-kinds=all

--92340-- --track-origins=yes

--92340-- --verbose

--92340-- --log-file=valgrind.log

--92340-- Contents of /proc/version:

--92340-- Linux version 5.15.167.4-microsoft-standard-WSL2 (root@f9c826d3017f) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP Tue Nov 5 00:21:55 UTC 2024

--92340--

--92340-- Arch and hwcaps: AMD64, LittleEndian, amd64-cx16-lzcnt-rdtscp-sse3-ssse3-avx-avx2-bmi-f16c-rdrand-rdseed-fma

```

--92340-- Page sizes: currently 4096, max supported 4096

--92340-- Valgrind library directory: /snap/valgrind/181/usr/libexec/valgrind

--92340-- Reading syms from /home/ameltonkhoza/COS214/COS214_Practical_5-
main/campusguard

--92340-- Reading syms from /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2

--92340-- Considering /usr/lib/debug/.build-
id/0d/77dbd4379bf0b5af3e05be7fb6af1e4cea7c96.debug ..

--92340-- .. build-id is valid

--92340-- Reading syms from /snap/valgrind/181/usr/libexec/valgrind/memcheck-amd64-
linux

--92340-- object doesn't have a dynamic symbol table

--92340-- Scheduler: using generic scheduler lock implementation.

--92340-- Reading suppressions file: /snap/valgrind/181/usr/libexec/valgrind/default.supp

==92340== embedded gdbserver: reading from /tmp/vgdb-pipe-from-vgdb-to-92340-by-
ameltonkhoza-on-???

==92340== embedded gdbserver: writing to /tmp/vgdb-pipe-to-vgdb-from-92340-by-
ameltonkhoza-on-???

==92340== embedded gdbserver: shared mem /tmp/vgdb-pipe-shared-mem-vgdb-92340-
by-ameltonkhoza-on-???

==92340==

==92340== TO CONTROL THIS PROCESS USING vgdb (which you probably
==92340== don't want to do, unless you know exactly what you're doing,
==92340== or are doing some strange experiment):
==92340== /snap/valgrind/181/usr/libexec/valgrind/../../bin/vgdb --pid=92340 ...command...
==92340==

==92340== TO DEBUG THIS PROCESS USING GDB: start GDB like this
==92340== /path/to/gdb ./campusguard
==92340== and then give GDB the following command
==92340== target remote | /snap/valgrind/181/usr/libexec/valgrind/../../bin/vgdb --
pid=92340
==92340== --pid is optional if only one valgrind process is running
==92340==

--92340-- REDIR: 0x4040b00 (ld-linux-x86-64.so.2:strlen) redirected to 0x58227ce6
(vgPlain_amd64_linux_REDIREC_FOR_strlen)

```

```

--92340-- REDIR: 0x403f4e0 (ld-linux-x86-64.so.2:strcmp) redirected to 0x58227d00
(vgPlain_amd64_linux_REDIREF_FOR_strcmp)

--92340-- REDIR: 0x403f2b0 (ld-linux-x86-64.so.2:index) redirected to 0x58227d39
(vgPlain_amd64_linux_REDIREF_FOR_index)

--92340-- REDIR: 0x403e640 (ld-linux-x86-64.so.2:bcmp) redirected to 0x58227d1b
(vgPlain_amd64_linux_REDIREF_FOR_memcmp)

--92340-- Reading syms from /snap/valgrind/181/usr/libexec/valgrind/vgpreload_core-
amd64-linux.so

--92340-- Reading syms from /snap/valgrind/181/usr/libexec/valgrind/vgpreload_memcheck-
amd64-linux.so

==92340== WARNING: new redirection conflicts with existing -- ignoring it

--92340-- old: 0x0403f4e0 (strcmp ) R-> (0000.0) 0x58227d00
vgPlain_amd64_linux_REDIREF_FOR_strcmp

--92340-- new: 0x0403f4e0 (strcmp ) R-> (2016.0) 0x04869780 strcmp

==92340== WARNING: new redirection conflicts with existing -- ignoring it

--92340-- old: 0x04040b00 (strlen ) R-> (0000.0) 0x58227ce6
vgPlain_amd64_linux_REDIREF_FOR_strlen

--92340-- new: 0x04040b00 (strlen ) R-> (2007.0) 0x04868530 strlen

--92340-- REDIR: 0x403e910 (ld-linux-x86-64.so.2:mempcpy) redirected to 0x486cff0
(mempcpy)

--92340-- Reading syms from /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33

--92340-- Reading syms from /usr/lib/x86_64-linux-gnu/libgcc_s.so.1

--92340-- Reading syms from /usr/lib/x86_64-linux-gnu/libc.so.6

--92340-- Considering /usr/lib/debug/.build-
id/a4/a7992a8e66555c8141ab2a08a8465ff6e0ea65.debug ..

--92340-- .. build-id is valid

--92340-- Reading syms from /usr/lib/x86_64-linux-gnu/libm.so.6

--92340-- REDIR: 0x4040ca0 (ld-linux-x86-64.so.2:strncmp) redirected to 0x4868fb0
(strncmp)

--92340-- REDIR: 0x4bdb150 (libc.so.6:strnlen) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdb1e0 (libc.so.6:stpbrk) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd92a0 (libc.so.6:strcmp) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bf24b0 (libc.so.6:wcsnlen) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

```

--92340-- REDIR: 0x4bd8390 (libc.so.6:memset) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bf1c20 (libc.so.6:wcslen) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4c5d650 (libc.so.6:___memcpy_chk) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd8300 (libc.so.6:memrchr) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bf2450 (libc.so.6:wcsncpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd7820 (libc.so.6:memcpy@@GLIBC_2.14) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bf09e0 (libc.so.6:wcschr) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd9190 (libc.so.6:index) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdb210 (libc.so.6:rindex) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bf0a90 (libc.so.6:wscmp) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd85b0 (libc.so.6:stpncpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bfefb0 (libc.so.6:wmemchr) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdaff0 (libc.so.6:strncmp) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd8610 (libc.so.6:strcasecmp) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bda410 (libc.so.6:strcspn) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bf19f0 (libc.so.6:wscpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd9120 (libc.so.6:strcat) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdaef0 (libc.so.6:strncasecmp_l) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd9210 (libc.so.6:strchrnul) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd7730 (libc.so.6:bcmp) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bda3a0 (libc.so.6:strcpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd86b0 (libc.so.6:strcasecmp_l) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdadc0 (libc.so.6:strlen) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdb090 (libc.so.6:strncpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bff030 (libc.so.6:wmemcmp) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4c5d770 (libc.so.6:___memmove_chk) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

==92340== WARNING: new redirection conflicts with existing -- ignoring it

--92340-- old: 0x04caed50 (___memcpy_chk_avx_una) R-> (2030.0) 0x0486d0f0
___memcpy_chk

--92340-- new: 0x04caed50 (___memcpy_chk_avx_una) R-> (2024.0) 0x0486ca60
___memmove_chk

--92340-- REDIR: 0x4bd8540 (libc.so.6:stpcpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd80c0 (libc.so.6:memmove) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

==92340== Preferring higher priority redirection:

--92340-- old: 0x04caed80 (___memcpy_avx_unalign) R-> (2018.0) 0x0486a8a0
___memcpy_avx_unaligned_erms

--92340-- new: 0x04caed80 (___memcpy_avx_unalign) R-> (2018.1) 0x0486c080
memmove

--92340-- REDIR: 0x4bd76b0 (libc.so.6:memchr) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdb3e0 (libc.so.6:strspn) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bf0420 (libc.so.6:wcpncpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bd81e0 (libc.so.6:mempcpy) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4bdae50 (libc.so.6:strncasecmp) redirected to 0x48551ba
(_vgnU_ifunc_wrapper)

--92340-- REDIR: 0x4cb3970 (libc.so.6:___strchr_avx2) redirected to 0x4867db0 (rindex)

--92340-- REDIR: 0x4cb1ac0 (libc.so.6:___strlen_avx2) redirected to 0x4868410 (strlen)

```

--92340-- REDIR: 0x4cae620 (libc.so.6: __memcmp_avx2_movbe) redirected to 0x486b7a0 (bcmp)
--92340-- REDIR: 0x4cb2ba0 (libc.so.6: __strncmp_avx2) redirected to 0x4868dd0 (strncmp)
--92340-- REDIR: 0x4bd3750 (libc.so.6: malloc) redirected to 0x485e799 (malloc)
--92340-- REDIR: 0x4caed80 (libc.so.6: __memcpy_avx_unaligned_erms) redirected to 0x486c080 (memmove)
--92340-- REDIR: 0x49358e0 (libstdc++.so.6: operator new(unsigned long)) redirected to 0x485ef23 (operator new(unsigned long))
--92340-- REDIR: 0x49338a0 (libstdc++.so.6: operator delete(void*)) redirected to 0x486280c (operator delete(void*))
--92340-- REDIR: 0x4caed40 (libc.so.6: __mempcpy_avx_unaligned_erms) redirected to 0x486cbf0 (mempcpy)
--92340-- REDIR: 0x4cb1310 (libc.so.6: __strcmp_avx2) redirected to 0x4869680 (strcmp)
--92340-- REDIR: 0x4bd3e30 (libc.so.6: free) redirected to 0x4861aba (free)

==92340==
==92340== HEAP SUMMARY:
==92340==    in use at exit: 0 bytes in 0 blocks
==92340== total heap usage: 205 allocs, 205 frees, 83,679 bytes allocated
==92340==
==92340== All heap blocks were freed -- no leaks are possible
==92340==
==92340== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)

```

The PDF should contain, in a sensible order:

- team member details;
- short system description;
- final UML class diagram;
- GoF participant mapping for all six patterns;
- justification for the two team-selected patterns;
- the three behavioural diagrams;
- concise design and ownership rationale;
- GDB investigation and Valgrind evidence;
- Dockerfile, docker-compose.yml, and the Docker run instructions used for the demonstration;
- GitHub repository link and short collaboration reflection; and
- a short contribution statement for each team member.